

A Experimental Set-up

A.1 Metrics

We use state-of-art performance metrics.

Performance metrics. Each classifier returns a prediction score S , where higher S values indicate a greater likelihood of $Y = 1$. This score is converted to a binary prediction using a threshold: $\hat{Y} = \mathbb{1}\{S > t\}$. Evaluating classifier performance with the prediction scores, unaffected by the threshold choice, is best measured by the Area Under the ROC Curve (AUC). The AUC score is given by:

$$P(S(x_i) > S(x_z) \mid y_i = 1, y_z = 0)$$

This formula measures the probability that the classifier ranks a randomly chosen positive instance higher than a randomly chosen negative instance.

We evaluate the performance of the default allocation and the fair allocation over a range of selection rates (where the number of selected instances is the same for both allocations). When the selection rate of a machine learning model is fixed, it implies that a set proportion of the instances will be chosen based on the highest scores predicted by the model, regardless of their actual scores. A useful performance metric to measure is the proportion of relevant instances among the top R instances selected by the model (where R denotes the resource level). This means the number of selected instances that actually have a positive target label (*True Positives* within the top R).

$$\text{Precision @}R = \frac{|\text{TP}_{\leq R}|}{R}$$

While precision at the top R relates to the efficient use of limited resources, knowing this value also determines the values of both recall and accuracy at R for a specific dataset and model and a fixed value of R [52]. We measure the cost of fairness as the loss in precision when using the fair allocation instead of the optimal allocation. For the sake of completeness, we also report the cost as loss in recall and loss in accuracy in Table 2 and Figures 3 and 4. The formula to calculate recall at the top R :

$$\text{Recall @}R = \frac{|\text{TP}_{\leq R}|}{|\text{TP}|}$$

The formula to calculate accuracy at the top R (with N being the size of the total population):

$$\text{Accuracy @}R = \frac{|\text{TP}_{\leq R}| + |\text{TN}_{>R}|}{N}$$

We focus on these widely used performance metrics, but the results hold for any weighted sum of costs of Type I/II errors.

Fairness Metrics. We focus on two standard fairness measures used to assess disparities between groups, with a sensitive attribute A .

Demographic parity (also known as statistical parity) requires that the rate of positive decisions is roughly equal for both the disadvantaged group and the advantaged group [21]:

$$P(\hat{Y} = 1 | A = s) \approx P(\hat{Y} = 1 | A = ns)$$

Equal opportunity requires the true positive rate to be approximately the same across groups [30], which enforces equal recall:

$$P(\hat{Y} = 1|A = s, Y = 1) \approx P(\hat{Y} = 1|A = ns, Y = 1)$$

B Results for other performance metrics

In this Section, we present the results for recall and accuracy. The general trends are in line with the results for precision.

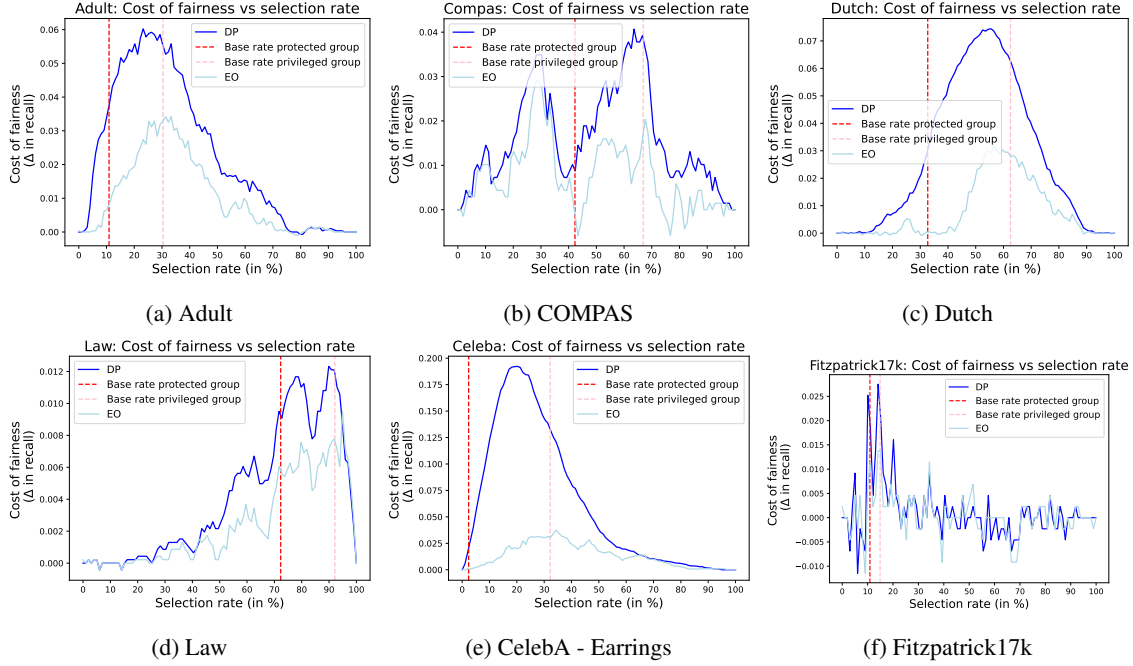


Fig. 3. Cost of fairness (loss in **recall**) for all datasets when enforcing demographic parity (DP) and equal opportunity (EO). The cost of fairness for both metrics heavily depends on the available resource level and thus the used selection rate.

Table 3. The average cost of fairness (the loss in recall and accuracy) when enforcing DP and EO. The average is calculated over all the selection rates from 1% to 100%.

Dataset	Adult	COMPAS	Dutch	Law	CelebA	Fitzpatrick17K
Avg. loss in recall (DP)	0.023	0.016	0.028	0.004	0.062	0.001
Avg. loss in recall (EO)	0.010	0.007	0.008	0.002	0.013	0.001
Avg. loss in accuracy (DP)	0.011	0.017	0.026	0.007	0.026	0.000
Avg. loss in accuracy (EO)	0.005	0.008	0.008	0.004	0.005	0.000

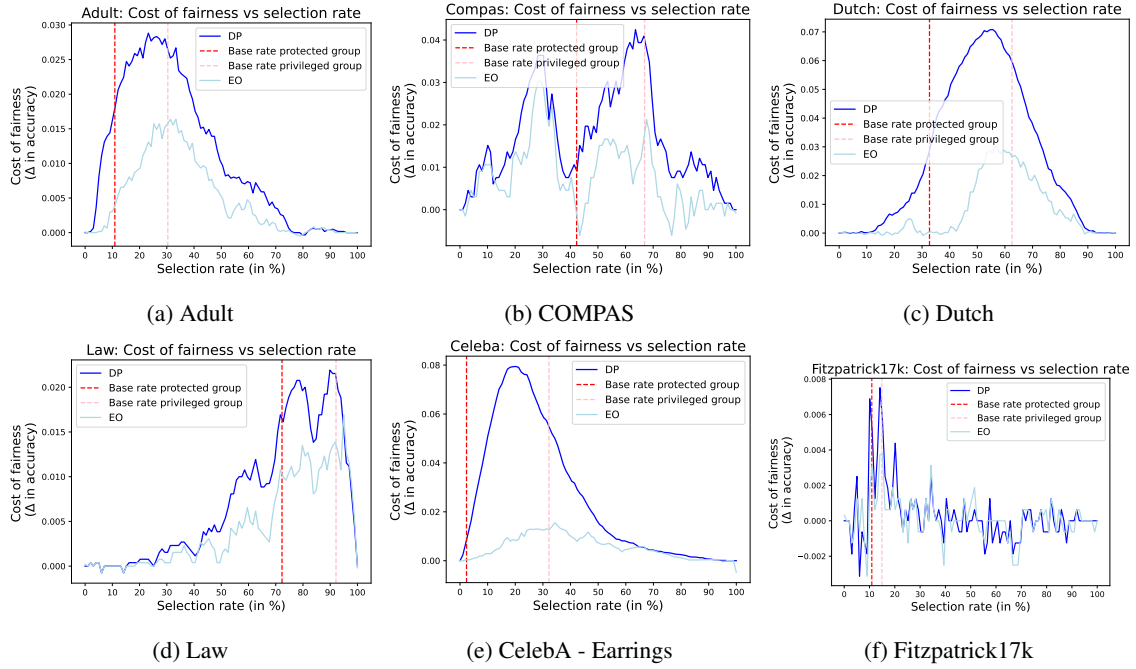


Fig. 4. Cost of fairness (loss in **accuracy**) for all datasets when enforcing demographic parity (DP) and equal opportunity (EO). The cost of fairness for both metrics heavily depends on the available resource level and thus the used selection rate.

C Illustration of other bias mitigation methods

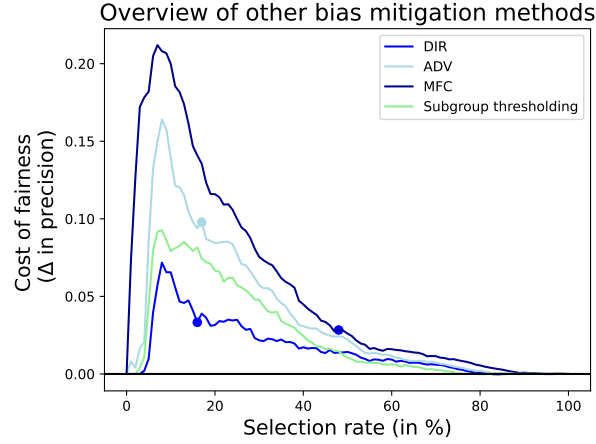


Fig. 5. Cost of fairness for different bias mitigation methods, namely the method we use (subgroup thresholding), DIR (Disparate Impact Remover [23]), ADV (Adversarial Debiasing [61]) and MFC (Meta Fair Classifier [11]). The dots indicate default thresholds.

Table 4. Model performance (entire population, advantaged group and disadvantaged group) and the average cost of fairness (loss in precision, accuracy and recall) when using the other bias mitigation methods compared to the performance when using the biased model. The average is calculated over all the selection rates from 1% to 100%. Results are calculated on the Adult income dataset. We report the AUC for every method separately as the prediction scores are also altered by the bias mitigation method. We also report the average demographic disparity and difference in equal opportunity as fairness is not enforced perfectly.

Bias mitigation method	Biased model	DIR	ADV	MFC
AUC	0.896	0.873	0.851	0.831
AUC_{priv}	0.887	0.866	0.845	0.816
AUC_{prot}	0.871	0.828	0.840	0.806
Avg. loss in precision	0.0	0.016	0.036	0.055
Avg. loss in recall	0.0	0.018	0.035	0.050
Avg. loss in accuracy	0.0	0.008	0.017	0.024
Avg. difference in DP	0.21	0.22	0.11	0.26
Avg. difference in EO	0.34	0.33	0.31	0.26

Although a systematic benchmark of bias mitigation methods is beyond the scope of this paper, we want to demonstrate that the core patterns observed in our analysis extend to a range of other bias mitigation methods as well. Therefore, we use three well-known bias mitigation methods: Disparate Impact Remover (DIR), Adversarial Debiasing (ADV) and the Meta Fair Classifier (MFC). We illustrate the results of this analysis in Figure 5 and

Table 4. Our theoretical reasoning does not hold for the behavior of preprocessing and in-processing bias mitigation methods, as the outcomes will depend on the specific characteristics of the underlying machine learning model. However, the main patterns we find hold true for these methods as well. Across all methods, we generally observe a bell-shaped curve, indicating that the cost of fairness is strongly influenced by the level of available resources (Figure 5). This can also mean that the cost can be heavily over- or under-exaggerated, and is highly context-dependent. We see that the used method in this paper (subgroup thresholding), has a cost curve that falls between those of the other mitigation techniques, while following a similar overall trajectory. Lastly, we can see that these methods do not enforce fairness perfectly (which will be the case for subgroup thresholding).

D Results for different resource levels

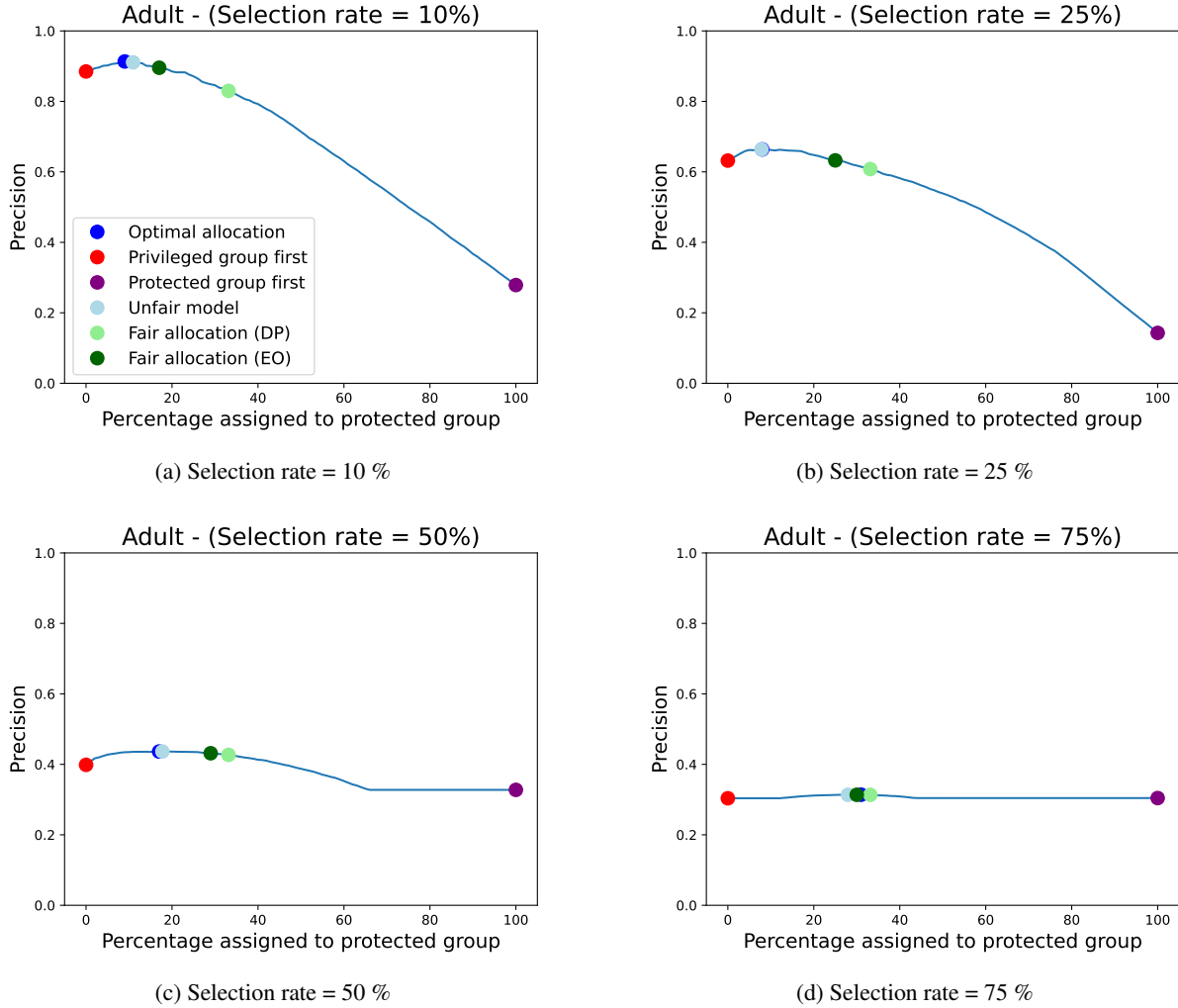


Fig. 6. Allocation results for different resource levels (Adult Income dataset). Left point on the x-axis represents the resources being maximally allocated to the advantaged group, while the right point represents the resources being maximally allocated to the disadvantaged group.

We study the effect of the different resource level on the allocation of resources. For the Adult dataset, this is shown in Figures 6a-6d. These figures demonstrate how different allocations lead to different levels of precision for one fixed resource level. The two extreme points of the curve represent all the resources being awarded to either the advantaged group (red dot on the left) or the disadvantaged group (purple dot on the right).¹⁰ We see that the

¹⁰However, also for these extreme points, when the total level of resources is higher than the size of one of the groups, some of the resources will still be awarded to the other group (as can for example be seen in Figure 6c).

optimum is not reached by awarding all the resources to one of the groups, but somewhere in the middle (dark blue dot). We note that the precision of the unconstrained (*'unfair'*) model, will be very close to the optimal allocation, and that the allocations required by the fairness metrics (both DP and EO) will lead to a lower precision for every resource level (Figures 6a-6d). However, this difference is a lot larger for low resource levels (Figures 6a-6b). In Figure 6c, the default allocation of the machine learning model is very unfair, but using the fair allocation (both DP and EO) results in only a little change in precision. This demonstrates that a very unfair initial model allocation does not guarantee a high cost of fairness. In Figure 6d, the default allocation of the machine learning model is already approximately fair. Hence, the cost of fairness is also low.